

HAIYANG DAO, China

WMO: 545870

Lat: 39.055N

Lon: 123.176E

Elev: 10

StdP: 101.20

Time Zone: 8.00 (E08)

Period: 90-04

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.7	(c) -8.1	(d) -20.4	(e) 0.6	(f) -7.5	(g) -18.4	(h) 0.7	(i) -6.1	(j) 18.5	(k) -4.2	(l) 15.8	(m) -5.7	(n) 8.4	(o) 0	(p) 0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.7	(c) 29.2	(d) 25.6	(e) 28.1	(f) 24.9	(g) 27.2	(h) 24.3	(i) 26.5	(j) 28.4	(k) 25.6	(l) 27.3	(m) 24.9	(n) 26.5	(o) 2.9	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.3	28.1	25.1	20.2	26.9	24.4	19.4	26.2	83.6	28.5	79.3	27.4	76.3	26.6	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 14.8	(b) 12.9	(c) 11.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 14.8	(b) 12.9	(c) 11.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	-1.9	0.2	4.1	10.0	14.9	20.0	23.6	24.9	21.6	15.0	7.4	0.6		
	DBStd	9.72	3.58	3.27	3.05	2.87	3.10	2.36	2.07	1.72	2.07	3.29	4.37	4.00		
	HDD10.0	1261	369	273	184	35	3	0	0	0	0	6	101	291		
	HDD18.3	2935	628	507	441	248	112	11	0	0	2	109	328	549		
	CDD10.0	1904	0	0	1	37	155	301	421	461	347	159	22	0		
	CDD18.3	538	0	0	0	0	6	62	163	203	99	5	0	0		
	CDH23.3	2414	0	0	0	0	5	120	769	1251	267	3	0	0		
	CDH26.7	316	0	0	0	0	0	3	108	183	22	0	0	0		
(14) Wind	WSAvg	4.7	5.7	5.0	5.3	5.2	4.5	3.5	3.9	3.2	3.5	4.7	5.7	5.7		
(15) Precipitation	PrecAvg	851	9	16	20	49	78	100	213	187	84	44	33	16		
	PrecMax	1156	26	45	98	97	175	189	477	344	201	81	76	47		
	PrecMin	602	1	3	1	7	20	21	55	58	19	6	4	5		
	PrecStd	155	6	11	18	24	37	42	78	79	43	22	21	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	9.1	14.2	18.9	24.2	27.1	30.5	31.0	28.6	23.3	17.7	10.9		
		MCWB	4.7	7.3	9.3	12.3	16.9	20.5	26.5	26.7	24.3	18.2	13.8	8.8		
	2%	DB	5.3	6.6	10.9	16.9	22.5	25.8	29.0	29.7	27.4	21.5	15.9	9.5		
		MCWB	3.5	3.9	7.5	11.0	16.2	20.0	25.1	26.4	23.3	17.5	13.3	7.8		
	5%	DB	4.2	5.5	9.4	15.7	21.3	24.7	27.8	28.6	26.1	20.3	14.8	8.2		
		MCWB	2.2	3.1	6.0	10.5	15.6	19.7	24.6	25.5	22.0	16.6	12.6	6.5		
	10%	DB	3.1	4.6	8.1	14.5	19.9	23.7	26.7	27.7	25.0	19.1	13.5	6.8		
		MCWB	1.2	2.5	5.1	9.8	15.0	19.3	24.1	24.9	21.0	16.1	11.2	4.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	8.0	10.4	14.7	18.6	22.8	27.6	28.4	25.3	19.7	15.2	9.4		
		MCDB	6.5	9.1	12.1	17.8	22.4	25.6	29.7	29.9	27.5	22.0	16.5	10.6		
	2%	WB	3.8	4.7	8.1	12.9	17.4	21.9	26.3	26.9	24.2	18.5	14.1	8.2		
		MCDB	5.1	6.0	10.4	15.5	20.7	24.3	28.2	28.8	26.6	20.5	15.6	9.3		
	5%	WB	2.5	3.7	6.7	11.6	16.6	21.1	25.2	26.1	23.1	17.6	12.8	6.7		
		MCDB	3.9	5.0	8.7	14.3	19.9	23.1	26.9	27.8	25.2	19.5	14.7	8.1		
	10%	WB	1.3	2.9	5.6	10.6	15.7	20.4	24.5	25.4	22.2	16.6	11.6	5.2		
		MCDB	2.8	4.4	7.6	13.4	19.1	22.4	26.0	27.1	24.2	18.9	13.3	6.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	5.0	5.0	5.8	5.7	5.2	4.4	4.7	5.6	5.8	5.6	5.1		
		MCDBR	6.2	5.5	6.0	7.1	6.8	6.8	5.8	5.7	6.8	6.8	7.0	6.0		
	5% WB	MCWBR	5.9	4.6	4.5	4.7	4.2	4.3	3.5	3.9	4.7	5.2	5.8	5.7		
		MCWBR	5.7	5.1	5.9	6.8	6.1	5.8	4.9	4.9	6.2	5.9	5.9	5.5		
(40) Clear-Sky Solar Irradiance	taub		0.429	0.496	0.547	0.620	0.648	0.735	0.715	0.615	0.541	0.565	0.516	0.443		
	taud		1.961	1.831	1.756	1.661	1.657	1.559	1.645	1.840	1.934	1.778	1.836	1.947		
	Ebn at Noon		688	692	706	686	679	621	629	683	706	618	587	633		
	Edh at Noon		133	174	207	242	247	273	249	200	171	181	151	126		
(44) All-Sky Solar Radiation	RadAvg		2.43	3.23	4.42	5.18	5.70	5.11	4.55	4.74	4.50	3.43	2.40	2.06		
	RadStd		0.18	0.25	0.33	0.39	0.47	0.57	0.47	0.51	0.35	0.23	0.24	0.14		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page